

AI-Based Smart Wireless Emergency Charging and Battery Health Prediction System

Final-Year Project Concept Document
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Project Theme: Wireless Power Transfer + IoT Battery Monitoring + AI/Data Science

1. INTRODUCTION

Portable electronic devices such as smartphones, sensors and emergency communication devices depend on rechargeable batteries. During emergencies, users may not have access to a suitable charging cable or power socket. Wireless power transfer provides a way to transfer electrical energy without a physical charging cable. This project extends that idea by adding sensors, a microcontroller and Artificial Intelligence/Data Science techniques for intelligent battery monitoring.

As an AI and Data Science project, the system does more than wireless charging. It collects battery data such as voltage, current, temperature, charging time and battery-cycle information. The data is cleaned, analyzed and used to train machine-learning models that can estimate State of Charge (SoC), predict battery health and identify abnormal battery conditions.

2. ABSTRACT

The proposed project develops a smart wireless emergency charging system that transfers electrical energy from a transmitter to a receiver connected to a rechargeable battery, eliminating the need for a physical charging cable during charging. Sensors continuously monitor important battery parameters such as voltage, current and temperature. The collected data is processed using Data Science methods including data collection, cleaning, feature analysis and visualization. Machine Learning is then used to estimate battery State of Charge, predict battery health and detect abnormal operating conditions. An ESP32 or similar microcontroller can collect sensor data and communicate with a monitoring dashboard. The project combines wireless power transfer, IoT, Artificial Intelligence and Data Science to create a safer and smarter emergency charging concept.

3. PROBLEM STATEMENT

- Conventional charging normally requires a physical cable and suitable charging port.
- Emergency situations may occur when a cable, adapter or power socket is unavailable.
- Users may not know the actual health or condition of a rechargeable battery.
- Continuous monitoring of voltage, current and temperature is often not available in simple charging systems.
- Abnormal temperature or charging behavior can reduce battery safety and lifetime.

Proposed problem solution: Develop a wireless emergency charging prototype with intelligent battery monitoring and AI-based battery health prediction.

4. OBJECTIVES

1. Develop a low-power wireless power transfer prototype.
2. Enable battery charging without a physical charging cable.
3. Measure battery voltage, current and temperature.
4. Collect and prepare battery data using Data Science techniques.
5. Estimate the battery State of Charge (SoC).
6. Predict battery health using Machine Learning.
7. Detect abnormal battery conditions.
8. Display battery information through a simple dashboard.
9. Demonstrate the integration of AI, IoT and wireless power technology.

5. EXISTING SYSTEM

Conventional charging flow: Power Source → Adapter → Physical Cable → Battery

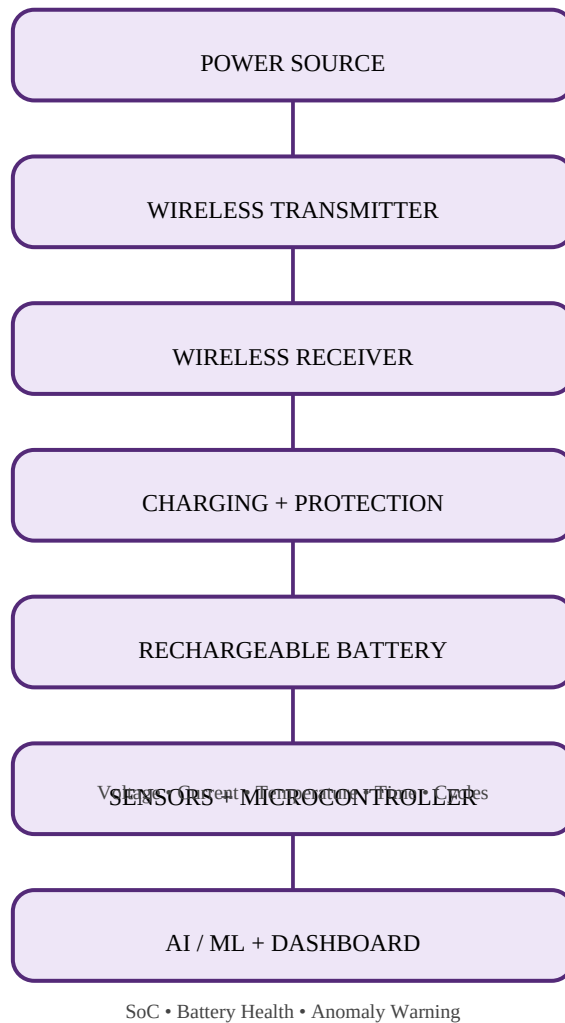
- Requires a physical charging cable.
- Charging port and cable can become damaged or worn.
- Less convenient when a cable is unavailable during emergencies.
- Basic systems may provide limited battery-health intelligence.
- Users may only see a battery percentage without deeper health analysis.

6. PROPOSED SYSTEM

The proposed system uses a wireless transmitter and receiver to transfer electrical energy without a physical charging cable. A charging and protection stage manages the battery. Sensors measure voltage, current and temperature, while a microcontroller collects the data. The Data Science pipeline cleans and analyzes the data, and a Machine Learning model predicts battery condition.

- Wireless power transfer for emergency charging.
- Real-time sensing of battery parameters.
- Microcontroller-based data acquisition.
- AI/ML prediction of battery health and charge level.
- Abnormal-condition warning.
- Dashboard for monitoring and visualization.

7. SYSTEM BLOCK DIAGRAM



Note: The wireless stage transfers energy from a real power source; the system does not create energy without an energy source.

8. METHODOLOGY

1. Study wireless power transfer, rechargeable batteries and Battery Management Systems.
2. Select a safe low-power wireless transmitter and receiver arrangement.
3. Connect the receiver to an appropriate charging and protection circuit.
4. Integrate voltage, current and temperature sensors with the microcontroller.
5. Collect charging and discharging data under controlled conditions.
6. Clean and preprocess the dataset using Python and Pandas.
7. Analyze features and identify patterns related to battery charge and health.
8. Train and test a Machine Learning model.
9. Connect predictions to a monitoring dashboard.
10. Evaluate wireless charging behavior, sensor readings and ML performance.

9. HARDWARE REQUIREMENTS

- ESP32 or Arduino-compatible microcontroller
- Wireless power transmitter module
- Compatible wireless power receiver module
- Rechargeable battery and suitable battery protection/charging module
- Voltage sensor
- Current sensor
- Temperature sensor
- OLED/LCD display
- LED/buzzer indicators
- Regulated power source
- Breadboard/PCB and connecting wires

10. SOFTWARE REQUIREMENTS

- Python
- NumPy and Pandas for data processing
- Scikit-learn for Machine Learning
- Matplotlib for visualization
- Arduino IDE for microcontroller programming
- Jupyter Notebook or Google Colab
- VS Code
- Optional: Streamlit for the monitoring dashboard
- Optional: Firebase/MySQL for data storage

11. AI / MACHINE LEARNING ALGORITHM

Recommended primary algorithm: Random Forest

Random Forest is a machine-learning method that combines multiple decision trees to produce a prediction. It is suitable for tabular battery data containing several input features such as voltage, current, temperature and charging-cycle information.

Input features: Voltage, current, temperature, charging time, discharging time and battery-cycle count.

Possible outputs: Estimated SoC, battery-health class/value and abnormal-condition status.

AI pipeline: Sensor Data → Data Collection → Data Cleaning → Feature Selection → Model Training → Testing → Prediction → Dashboard

Example: Voltage = 3.85 V, Current = 0.45 A, Temperature = 31°C, Cycles = 120 → ML Model → Predicted Battery Health = 87% → Status = GOOD.

Possible comparison algorithms: Decision Tree, Linear Regression and XGBoost.

12. ADVANTAGES

- Charging without a physical cable connection.
- Useful as an emergency charging concept.
- Real-time monitoring of important battery parameters.
- AI-based battery health prediction.
- Potential early warning of abnormal battery behavior.
- Data-driven analysis suitable for an AI/Data Science project.
- Combines hardware, IoT, AI and Data Science in one system.

13. FUTURE SCOPE

- Improve wireless charging efficiency and charging distance.
- Develop faster wireless power transfer methods.
- Use AI to optimize charging based on battery condition.
- Predict remaining battery useful life.
- Support multiple compatible devices.
- Combine renewable sources such as solar energy with wireless charging.
- Develop smart emergency charging stations for public and disaster-response environments.
- Integrate the technology into future IoT devices, vehicles and portable systems.

14. EXPECTED RESULT

- A working low-power wireless charging demonstration.
- Real-time sensor readings for voltage, current and temperature.
- A prepared battery dataset for AI/Data Science analysis.
- An ML model capable of estimating battery condition from measured data.
- A dashboard showing charge level, health status and warnings.
- Demonstration of an integrated AI + IoT + wireless power prototype.

15. CONCLUSION

The AI-Based Smart Wireless Emergency Charging and Battery Health Prediction System combines wireless power transfer with intelligent battery monitoring. The wireless section removes the need for a physical charging cable in the prototype, while sensors provide real-time battery data. As an AI and Data Science project, the collected data is cleaned, analyzed and used for Machine Learning-based prediction of battery charge level, battery health and abnormal conditions. The project therefore demonstrates a practical way to combine emerging wireless-power technology with AI-driven battery intelligence for safer and smarter emergency charging.

THANK YOU

Questions?